



Rytha Mitra

Intelligent Farming Companion for Every Farmer

 WitchHunt 2026 • Climate Action Track



Harvest Hex Harvesters



MEET THE TEAM



AI4India



Yashaswini V

Data Science & AI/ML

Passionate about building intelligent solutions that create real impact in rural communities. Enthusiastic open source contributor and lover of ML, NLP & LLMs.

 Open Source Contributor

Skills:

Random Forest

SHAP

CrewAI

LLM Orchestration



Darshini K.H

Full Stack Developer

Loves turning ideas into reliable, user-friendly products that work anywhere. Focused on building fast, resilient and offline-first web experiences. Interested in earning and problem solving.

Skills:

Flask API

Frontend (HTML/CSS/JS)

PWA

Offline-first



Our Hackathon Journey

From understanding farmers' real challenges in **Karnataka** to building an AI-powered solution that works even without internet — this journey is about impact, learning and building for the future.



Farmer-first

Solutions built for real farmers



Data-driven

Insights powered by real-world data



100% Offline-capable

Works anywhere, anytime



Built for Karnataka

Designed for local needs, by local minds



OPPORTUNITY IN RURAL ADVISORY

A big gap.
A **bigger opportunity.**
A better future for every farmer.



Huge unmet need

for localized, real-time crop and market guidance in Karnataka.



Farmers are ready

for simple, trusted, mobile-first tools that save time and reduce risk.



AI, voice, and local datasets

now make advisory services scalable and affordable.



Rythra Mitra

turns this gap into an opportunity for better crop decisions, better market timing, and better income.



Speaker Line

“ Instead of only showing the pain, we wanted to show the gap we can solve:
a **large opportunity for trusted, explainable, farmer-first advisory.** ”





AI4India

The Complete Farming Intelligence Platform

An AI-powered, space-tech platform that helps farmers plan, protect, and grow – with confidence.

All-in-one. Data-backed. Farmer-first.



Strategic Crop Planning

AI (Random Forest) + SHAP explainability for smarter, data-driven crop decisions.



Instant Pest Diagnosis

Vision AI identifies pests & diseases in seconds and recommends targeted solutions.



Field-Level Monitoring

NDVI satellite data to track crop health, soil moisture, and stress – every field, every time.



Farm Credit Score

AI-powered credit assessment to help you access loans faster and grow bigger.



Hyper-Local Intelligence

Real-time mandi prices, weather alerts, drought warnings & local advisory in your language.



AI + ML
Powered



Space-Tech
Insights



Trusted by
Farmers



For Farmers.
By India.



Try Now →

Click on the mobile. Start your smart farming journey today!



FARMER PERSONA



Name

Ramesh, 42



Role

Progressive farmer



ABOUT RAMESH

Ramesh grows multiple crops and wants better planning for sowing, pest control, and selling at the right time. He is open to technology but needs fast, practical guidance.



PAIN POINTS



Market uncertainty



Crop loss risk



Advice that is too generic



“ I want the right advice, at the right time, so I can protect my crops and get the best price for my produce. ”



FARMER PERSONA

 Name	 Role
Lakshmi, 34	Smallholder farmer

ABOUT LAKSHMI

Lakshmi manages a small family farm in Karnataka. She wants timely crop advice, weather updates, and market price alerts in Kannada.

FRUSTRATIONS

-  Delayed information
-  Hard-to-read apps
-  Dependence on others for decisions



“ I just want the right information, at the right time, in my language. So I can take the right decision for my farm and my family. ”



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COMMUNITY MEMBER / OWNER PERSONA



Role

Village cooperative member /
community owner



ABOUT THE COMMUNITY

Community members and owners work together to coordinate purchases, share information, and make group decisions. They want a tool that can support the whole community, not just one farmer.



FRUSTRATIONS



Lack of
centralized
data



Poor
communication



Delayed
updates



“ When the whole community grows together, everyone benefits. I need a tool that gives us the right information, at the right time. ”



IMPACT

Early Impact Signals (Pilot: 10+ Users)



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1 PILOT REACH



10+ pilot users
from Karnataka
farming context



Tested in real scenarios
Kannada-first advisory
tested in real decision
scenarios



Mobile-first usage
No app install
required

2 WHAT USERS SAID



Positive feedback
on clarity and ease of
understanding



Faster decisions
compared to generic
advisory messages



Users reported confidence
in trying recommendations
for next cycle

3 WHAT CHANGES NEXT



Expand reach
from 10+ to 50+ users



**Add structured
pre/post measurement:**
water, fertilizer cost,
crop-confidence



Partner validation
with local extension/NGO
ecosystem

“

ಸಲಹೆಗಳು ನನ್ನ ಭಾಷೆಯಲ್ಲಿ ಅರ್ಥಮಾಡಿಕೊಳ್ಳಲು ಸುಲಭವಾಗಿದೆ.
ಇದು ನನ್ನ ಬೆಳೆಯ ನಿರ್ಧಾರ ತೆಗೆದುಕೊಳ್ಳುವಲ್ಲಿ ಸಹಾಯ ಮಾಡಿದೆ.

— ಪೈಲಟ್ ಬಳಕೆದಾರರು, ಮಂಡ್ಯ ಜಿಲ್ಲೆ



Current stage: Early qualitative + usage validation (hackathon pilot),
with clear path to quantified seasonal impact.





Our Journey Ahead

From a powerful MVP to a nation-scale farming intelligence ecosystem.



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PHASE 1 — MVP (COMPLETED)



MVP Launched & Live

Delivering real value to farmers today.



Data

Synthetic data (2.2K rows) covering 9 districts



Model

Random Forest (RF)



Explainability

SHAP explanations for every recommendation



Core Engine

100% Offline-First Engine (PWA + Local JS)



Voice Interface

Kannada TTS (Sarvam AI) – No typing needed



Performance

91.7% Accuracy (Validated)



Cost

₹0 Stack Cost (100% Open Source)



What's Live Today?

- ✓ Crop Recommendation
- ✓ Pest & Disease Diagnosis
- ✓ Weather & Drought Alerts
- ✓ Field NDVI Mapping
- ✓ Mandi Prices
- ✓ Farm Credit Score



PHASE 2–4 (NEXT 6 MONTHS)



Scaling Impact

Expanding depth, accuracy & reach across India.

PHASE 2

Month 1–2



Real Farmer Data (10K+ rows and growing)



Model Upgrade: XGBoost



Accuracy Target: 95–97%



More Districts: 15+ Districts

PHASE 3

Month 3–4



Seasonal & Regional Models



Yield Prediction Engine



IoT Sensor Integration



More Districts: 25+ Districts

PHASE 4

Month 5–6



Multi-language Support (Kannada, Telugu, Marathi, Hindi)



Community & Cooperative Dashboard



More Districts: 30 Districts (Pan-India Coverage)



Data

10K+ Real Rows



Accuracy

95–97%



Reach

30 Districts



Languages

4 Languages



Our Vision

Empowering every farmer with trusted, intelligent and accessible farming advice.



Built for Trust

Transparent. Ethical. Farmer-first.



Built for Bharat

Offline-first. Local-first. Bharat-first.



Built Together

With farmers, for farmers. Always.



Built to Scale

From villages to India. From today to the future.



Competitor Analysis & Our Moat

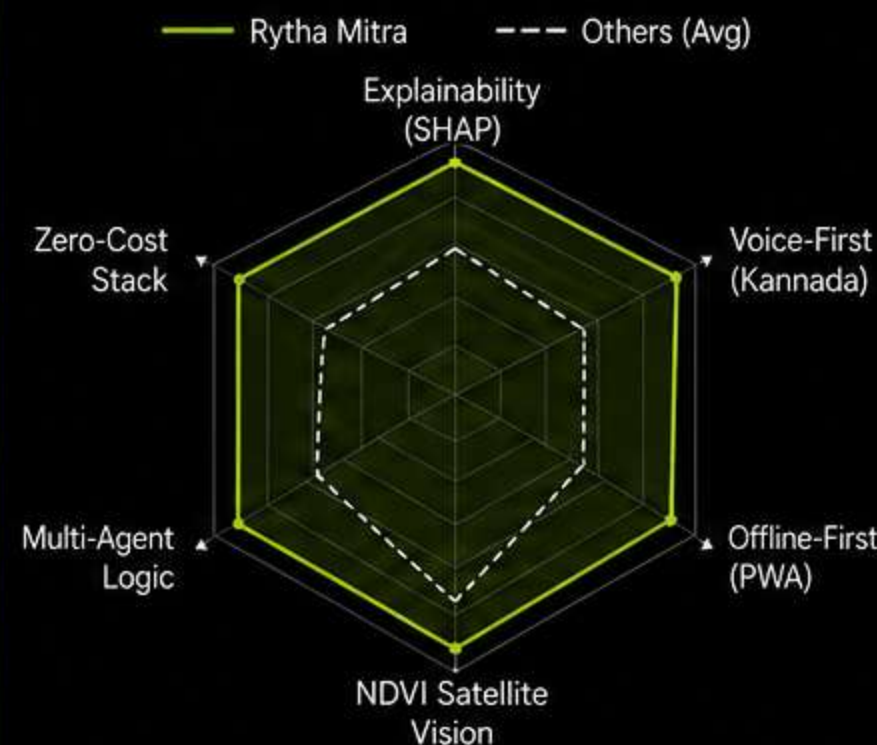
Rytha Mitra is built for Bharat. By Bharat.



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Feature	Rytha Mitra	plantix	AgroStar	DeHaat	Kisan Suvidha
Explainable AI (SHAP)	Yes	No	No	No	No
Voice-First (Kannada)	Yes	Basic	No	No	No
Multi-Agent Logic	Yes	No	No	No	No
Offline-First PWA	Yes	No	No	Partial	No
NDVI Satellite Vision	Yes	No	Beta	No	No
Zero-Cost Stack	Yes	Paid	Paid	Paid	Yes

1) Capability Radar



2) Our Strongest Moat



Explainable AI
Why, not just what.



Voice + Offline First
Built for Bharat's real connectivity.



4-Agent Intelligence
Science panel mimicry.



NDVI + Field Insight
See what you can't see.



Zero-Cost Stack
Sustainable & scalable for all.

Why it matters



Explainability (SHAP) → builds farmer trust and enables transparent extension services.



Voice + Offline → increases adoption in low-connectivity, Kannada-first user segments.



Multi-agent + NDVI → delivers richer, context-aware recommendations beyond single-model outputs.



Risk Mitigations



Risk: Competitors may add features quickly
Mitigate: open-data partnerships, faster field pilots.



Risk: External APIs dependency (weather/market)
Mitigate: cached historical fallbacks + offline heuristics.

Rytha Mitra Difference



Most agritech → dashboards



SHAP explainability (why, not just what).



Most require internet



100% offline via PWA + local JS engine.



Most English-first



Kannada voice (Sarvam AI), no typing.



Most single-model



CrewAI 4-agent debate (science panel mimicry).



Most fragmented



Unified ecosystem (diagnostics + monitoring + finance).



Trust & Transparency

SHAP explainability builds trust with farmers and extension officers.



Inclusivity & Accessibility

Voice-first Kannada + Offline-first design ensures no farmer is left behind.



Impact at Scale

Zero-cost, scalable, and built for Bharat — empowering 65 lakh+ farmers and beyond.

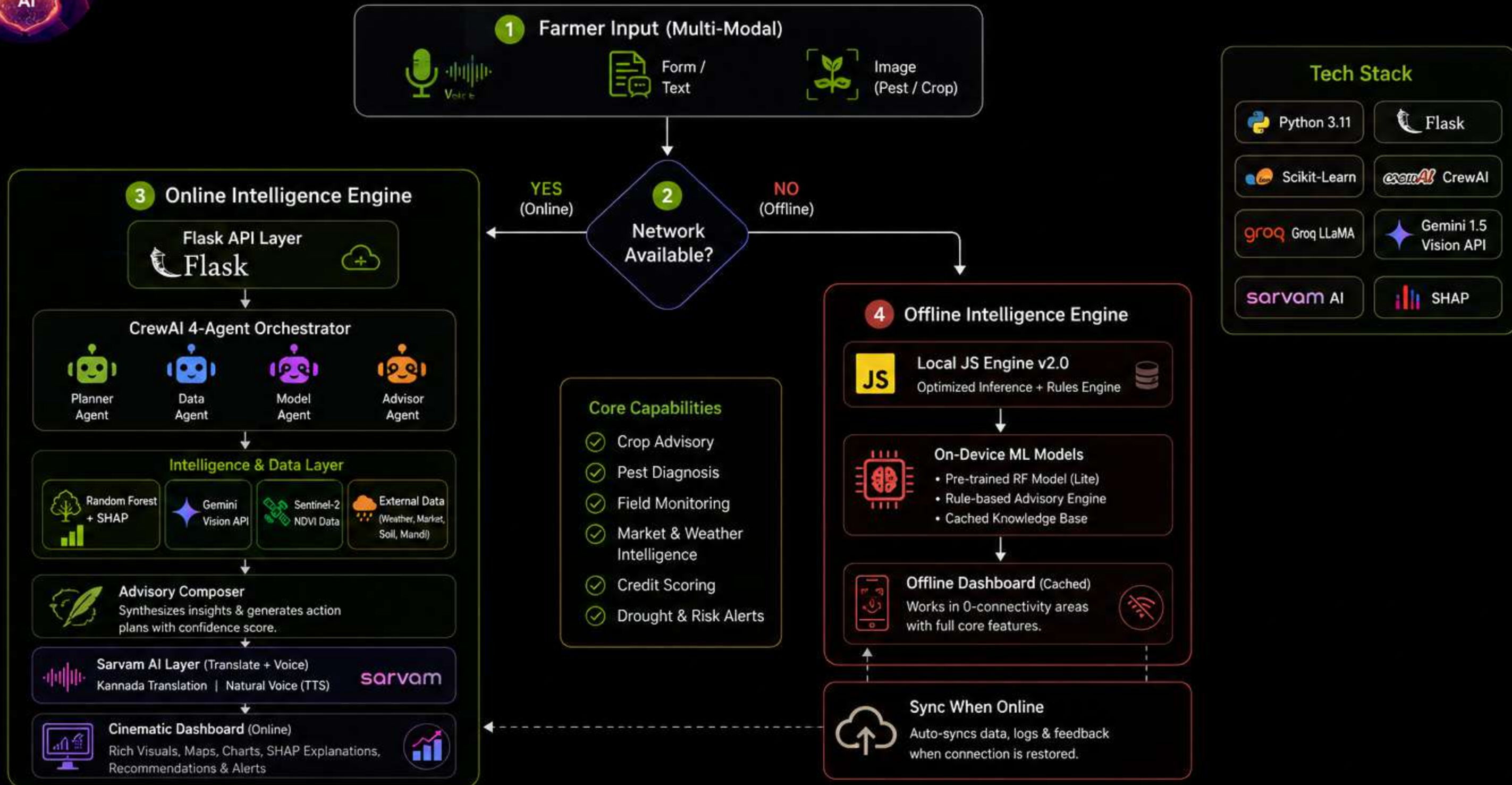


Agri Intelligence System Architecture

Dual-Engine. Always-on Intelligence for Every Farmer.



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Tech Stack

Python 3.11

Flask

Scikit-Learn

CrewAI

Groq LLaMA

Gemini 1.5 Vision API

sarvam AI

SHAP

Why This Architecture is Powerful



Dual-Engine Genius

Works online with full power, offline with intelligence.



Farmer First

Always available, always understandable.



Explainable AI

SHAP ensures transparency & trust.



Future Ready

Modular, scalable & ready for nationwide impact.

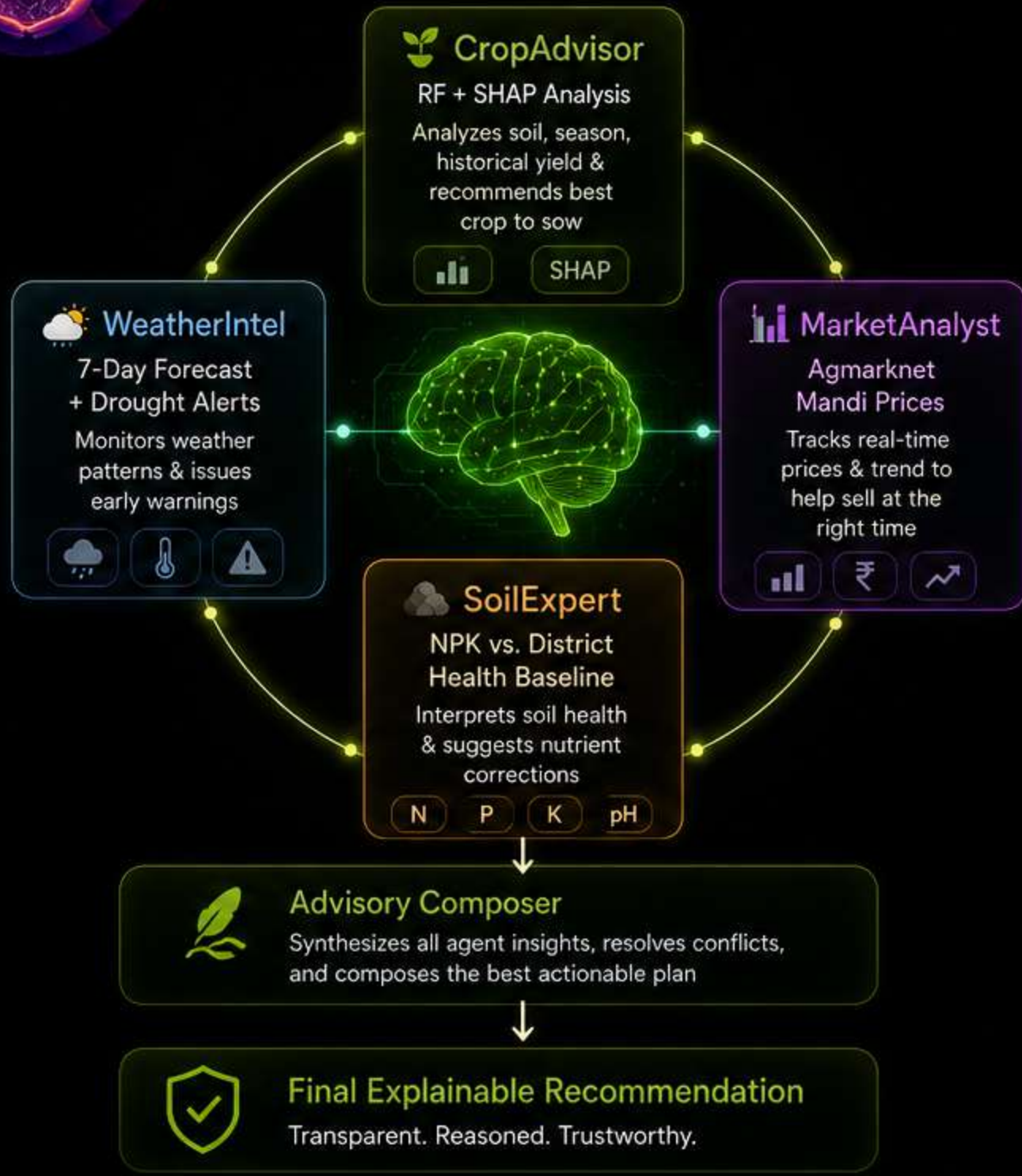
Built for Bharat

0-connectivity. Multi-language. Voice-first. Farmer-owned.



The 4-Agent Brain (CrewAI)

Farmer-Level Intelligence via Scientific Panel Debate



Process & Data Flow

End-to-End Intelligence Journey for Every Farmer



Performance Assurance

 < 2s

Online Latency

 < 500ms

Offline Latency

Always fast.
Always reliable.



Always Farmer-First.



Inclusive by Design: Karnataka First, India Next



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Current Phase (Now)



Geography

Karnataka-focused deployment



Language

Kannada-first advisory output



Accessibility

Mobile browser, low-bandwidth friendly, simple UI



Fairness

District-aware recommendations and confidence-aware outputs



Explainability

SHAP reasons + weather/soil rationale shown to user



Expansion Phase (Coming)



Add multilingual support in phases



Retain explainability and fairness checks across language rollouts

SHAP reasons



Fairness check



Scale district model to more states after Karnataka validation



Key Strategy

Depth-first rollout: maximize trust and adoption in one region/language, then **scale responsibly**.





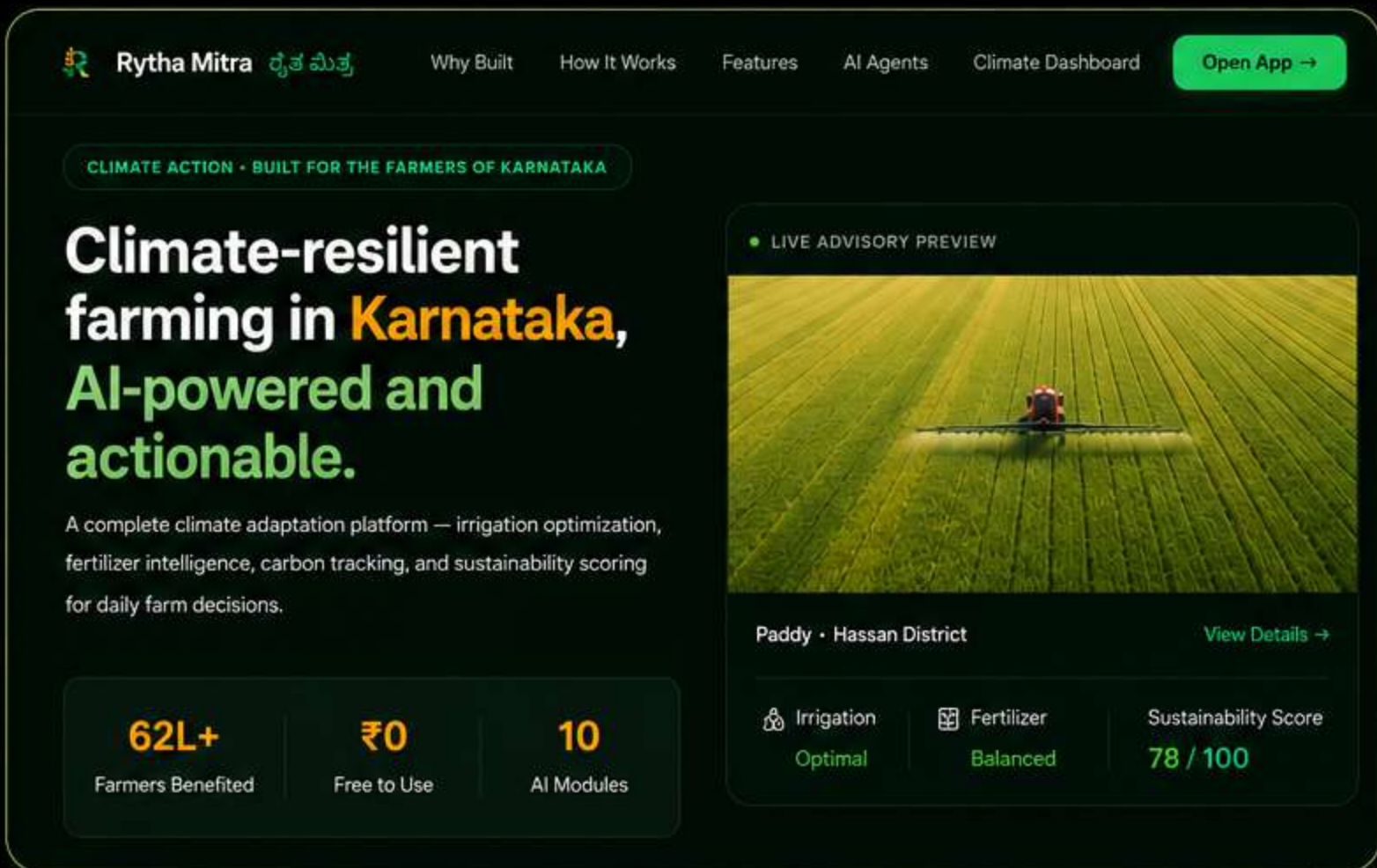
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MVP Screenshots

1 Splash Screen



2 Landing Page





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MVP Screenshots

3 Agricultural Intelligence Center

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← Landing Advisory Tool **Climate Dashboard** Simulator

ENTERPRISE AI PLATFORM • GROO LLAMA 3.3 • SHAP • SARVAM AI

Agricultural Intelligence Center

Real-time climate resilience engine for Karnataka's agrarian future.

Crop Advisor Market Analyst Weather Intel Soil Expert IN Kannada Output

2200
Training Samples

100
RF Trees

4
AI Agents

₹0
Stack Cost

14
Districts

CropAdvisor
RF + SHAP → top 3 crops

MarketAnalyst
Agmarket → profit

WeatherIntel
OWM 7-day flag

SoilExpert
NPK alerts

IN Sarvam AI
Kannada output

SEASONAL TIPS
SOWING WINDOW
Rabi: Oct-Nov • Kharif: Jun-Jul
IRRIGATION
Water at 5-7 AM. Skip if rain forecast >4mm

4 Climate Intelligence Dashboard

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← Home Advisory Tool **Agricultural Simulator** Irrigation Engine District Map

CLIMATE RESILIENCE PLATFORM • WITCHHUNT 2026

Climate Intelligence Dashboard

Enterprise-grade decision intelligence for climate-resilient agriculture in Karnataka. Real logic. Explainable outputs. Zero cost.

2400
LITERS WATER SAVED/ACRE

180
KG CO₂ REDUCED

72
SUSTAINABILITY SCORE

14
DISTRICTS COVERED

Launch Simulator →

Irrigation Engine

District Map

PLATFORM MODULES Irrigation Engine Fertilizer AI Climate Crops Carbon Tracker Sustainability Score District Intel Map



AI4India

THANK YOU

Together, we're building a future where
AI and **agriculture** grow together —
for people, for planet, for tomorrow. 🌱



GitHub
Yashaswini-V21



Live URL
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